

The Proving Line

京张验线

A Conceptual Urban Design Proposal for the Centennial Jing-Zhang AI Innovation Belt

A century ago every section of the Jing-Zhang line had to pass track-proving acceptance before opening; a century later this line is where global AI capabilities are proved before entering the real city — proved first, then opened, with rollback available and retirement possible.



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Iteration / Date	v1.0 · 2026-08-08
Tracks	ai-origin-community / jingzhang-heritage-narrative / civic-agent-governance
Format	A3 landscape booklet 420×297 mm · 13 pages

AI-generation disclosure: produced by an AI agent (Claude Code) under the human participant's authorization; every factual statement is linked to public authoritative sources (see sources.json).

Provisional Boundary note: all spatial drawings derive from PROVISIONAL rough boundaries supplied by the organizer, not Official Planning Boundaries; everything herein is a Conceptual Recommendation — it does not replace statutory planning nor constitute any approved conclusion.

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How This Booklet Maps to the Package

- proposal.md is the authoritative text: every number and phrase here matches it verbatim — this booklet only condenses layout
- metrics.json is the single source of values; geometry/*.geojson the source of spatial layers
- The three matrices (compliance / standard / depth) plus sources.json form the machine-readable evidence chain (P12)

Team & Agent Statement

- Participant: gehryliuRMuniversity (human — authorization & final review)
- Agent: Xiaoming (built on Claude Code) — research, design generation, drawing assembly and machine-readable self-checks
- Human-AI split: the agent generates and logs everything; the human authorizes submission; formal effect rests with official review and statutory procedures
- License: COMMUNITY-DISPLAY-ONLY
- Tracks: ai-origin-community / jingzhang-heritage-narrative / civic-agent-governance
- Scenarios: 13 cards incl. cultural guide, walkability, robot delivery, enterprise copilot, health navigation, expo roadshow — 13 cards (P11)

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Booklet structure follows Section 8 of the official guide: basis (P3) — scope (P4) — diagnosis (P5) — concept (P6) — systems (P7-9) — key areas (P10) — scenarios & operations (P11) — metrics & compliance (P12) — risks & gaps (P13).

All 27 sources are registered in sources.json and used by grade: formal basis sets task calibers; site facts anchor the scheme in reality; provisional boundaries serve generation and self-checks only; global cases lend mechanisms only. Standards base: MOHURD urban-design and regulatory-planning measures plus the MNR land-use classification guide.

Tier	ID	Title	Publisher
Formal basis	DATA-SRC-OFFICIAL-ANNOUNCEMENT-20260509	Prequalification Announcement, Intl. Solicitation for the Belt	Haidian Sub-bureau, BJ Planning & NR Commission
Formal basis	DATA-SRC-AGENT-TASKBOOK-20260518	Open Call Taskbook for Global Agents (cleared excerpt)	Zhongguancun Science City Admin Committee
Provisional boundary	DATA-SRC-PROVISIONAL-BOUNDARIES-20260605	Provisional rough boundary data	open-city.ai maintainers
Provisional boundary	SITE-PACKAGE	Structured site package (enums / ranges / standards)	open-city.ai
Site facts	SRC-PARK-PHASE2-OPENING	Heritage Park Phase 2 opens (9 km / 53 ha / 70 communities)	Qianlong.com
Site facts	SRC-PARK-PHASE1	Park Phase 1 completed (2.4 km / 16.8 ha)	BJ Gardening & Greening Bureau
Site facts	SRC-QINGHUAYUAN-STATION	Qinghuayuan Station restoration & exhibition	China News Service
Site facts	SRC-HAIDIAN-AI-BASELINE	Haidian AI industry baseline (March 2026)	Xinhuanet
Site facts	SRC-AI-ORIGIN-COMMUNITY	AI Origin Community progress (300+ AI firms, >70% share)	Beijing Daily
Site facts	SRC-HAIDIAN-POLICY	Haidian AI policy & Two-Zones-One-Belt layout	Haidian District Government
Site facts	SRC-JZ-HISTORY	Jing-Zhang Railway history, 1905-1909	Beijing Daily
Site facts	SRC-JZ-HERITAGE-NATIONAL	National heritage listing, Nankou-Badaling section	BJ Cultural Heritage Bureau
Site facts	SRC-HERITAGE-TEMPLES	Juesheng (Big Bell) & Zhenjue temples heritage info	Beijing Municipal Gov portal
Site facts	SRC-METRO-STATUS	Line 13 split; Line 18 first section opening	Xinhuanet
Site facts	SRC-CPX-NANYAN	Changping Line south extension through (Dec 2024)	Beijing Daily
Site facts	SRC-DZS-RENEWAL	Dazhongsi retail closure-renovation starts (Apr 2025)	The Beijing News
Site facts	SRC-METRO-18-OPENING	Metro Line 18 (Malianwa-Tiantongyuan E.) in trial, 2025-12-27	Beijing Daily (jingbao.net)
Site facts	SRC-XYL-INNOVATION	Xueyuan Rd 'Three Belts, One Festival' innovation practice	CAU News (reposted media)
Global cases	SRC-CASE-STATIONF	Station F, Paris	Station F
Global cases	SRC-CASE-KENDALL	Kendall Square innovation-space zoning	Cambridge Redevelopment Authority
Global cases	SRC-CASE-ONENORTH	one-north LaunchPad decade results	JTC
Global cases	SRC-CASE-KINGSCROSS	King's Cross development overview	King's Cross Central LP
Global cases	SRC-CASE-PANGYO	Pangyo Techno Valley land mechanism	Invest Korea
Global cases	SRC-CASE-HETAO	Hetao SZ-HK cooperation zone plan & data pilot	Shenzhen DRC
Global cases	SRC-CASE-TORONTO	Toronto Quayside termination (cautionary case)	CBC News
Global cases	SRC-CASE-KASHIWA	Kashiwa-no-ha tri-party governance & area energy	Mitsui Fudosan / CLC
Global cases	SRC-TORONTO-CLOSURE	Sidewalk Labs abandons Quayside (2020-05-07)	Sidewalk Labs (Medium)

Three-Level Scope Framework

- Coordinated Research Area (CRA), c.43.6 km2: to the 5th Ring (N), Jingzang Expwy (E), Xizhimenwai St (S), Wanquanhe Rd (W) — how the proving regime supports a world-class AI ecosystem
- Overall Design Area (ODA), c.11.4 km2: the 1-2 km urban band around the heritage park; urban renewal plus regulatory-depth urban design (land use / mobility / blue-green / character)
- Key-Area Detailed Design Area (KDA), c.368.4 ha: Zhongzhiyuan c.192.1 ha, AI Origin Community c.104.3 ha, Dazhongsi c.72.0 ha — implementation-plan design depth
- Cascade: CRA sets the ecosystem, ODA sets the structure, KDA sets implementation

Notice: Provisional Rough Boundary

The site boundary and all three key-area polygons are PROVISIONAL rough boundaries (official_boundary=false / geometry_role=provisional_constraint) for generation, visualization and self-checks only — never official red lines, approval bases or precise area references. Cited areas follow the announcement; once official red lines are issued, all land-use shares, green and public-space ratios, key-area layouts, phasing and every drawn number must be recomputed.

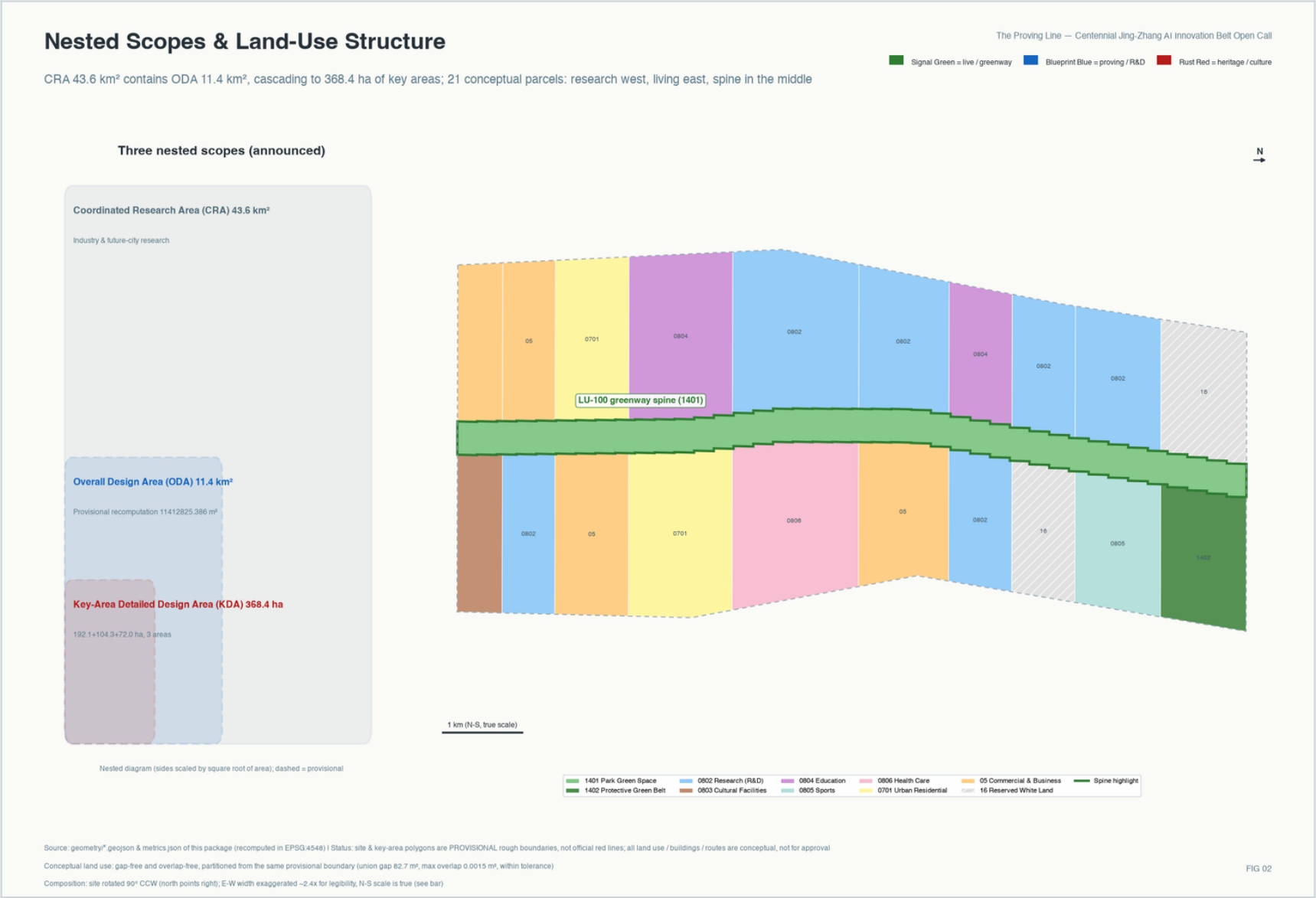


Fig.: scope cascade & land-use structure (from assets/figures/)

Anchors of Existing Conditions

- Heritage Park Phase 2 opened 2026-08-06; with Phase 1 it forms a 9 km, c.53 ha cultural green corridor from Xizhimen to the North 5th Ring, serving 70 communities and c.450,000 residents
- Phase 1 (2.4 km, c.16.8 ha) completed June 2023
- Qinghuayuan Station: built 1910, restored & reopened 2023; dual identity of railway heritage and revolutionary relic
- Dazhongsi's big-box furniture retail began closure-renovation in 2025 — one of the belt's most representative whole-building malleable spaces today
- Rail: Line 13 split advancing; Line 18 (Malianwa-Tiantongyuan East) in trial operation since 2025-12-27; Changping extension through since Dec 2024
- Haidian AI baseline (March 2026): 2,000+ AI firms, 26 unicorns, 130 registered LLMs, core industry above 350 billion yuan, 95,000 professionals, 135 AI2000 top scholars
- District policy sets the 'Two Zones, One Belt' AI innovation layout — this belt is the 'One Belt'

Five Severance Points = Five Stitching Corridors



Four Real Urban Questions This Scheme Answers

(1) For a century the rail corridor has drawn a north-south wall through the city, with campuses, parks and communities turning their backs on each other; (2) the c.450,000 residents of 70 communities need tangible service gains and the right to know, not abstract industry narratives; (3) this is fully built-up land — every ambition must be realized within existing stock; (4) for AI to enter public space, the first threshold is trust, not technology. The proving regime answers all four: five corridors re-stitch east and west, the four steps earn trust under public watch, stock activates as proofs pass, and the greenway returns gains to residents.

Issue List

Issue	Diagnosis & response
East-west severance	The rail corridor has long severed the city at five points — Dazhongsi, Xitucheng, Wudaokou, Qinghua East Rd, Zhongzhiyuan — the five stitching corridors of this scheme
Missing regulatory controls	FAR / height / coverage / setbacks unpublished — intensity follows the formula-only discipline; no invented numbers
Missing official red lines	Official red lines not public — all layers rest on provisional boundaries; release triggers full recomputation
Missing baselines	Building stock, tenure, heritage control lines, municipal and service capacities absent (nine gap classes, P13)
Operations gap	Park hardware is complete; renewal shifts from construction to AI narrative layering and scenario operations

Diagnosis: hardware (park and rail) is ready; the deficit is institutional and operational — precisely why the Proving Line treats institutions as the core design object.

Concept: prove first, then open, with rollback and retirement

The 'Proving Line' translates the segment-by-segment acceptance tradition of the Jing-Zhang Railway (begun 1905, opened 1909) led by Zhan Tianyou — trust earned through proof. The scheme upgrades the 9 km heritage corridor into the proving line global AI must pass before entering the real city: one line, three proving districts, five stitches, two wings.

- One line: the 9 km proving greenway on the completed park spine
- Three districts: Heavy-Proving (Zhongzhiyuan) / First-Proving (AI Origin Community) / Go-Live (Dazhongsi)
- Five stitches: E-W corridors at Dazhongsi, Xitucheng, Wudaokou, Qinghua East Rd, Zhongzhiyuan
- Two wings: Zhongguancun Technology Services Wing (W — depot & supply) + Xiaoyue River Scenario Enablement Wing (E — open field)
- Three positionings: centennial culture belt / urban AI life experience belt / AI innovation belt
- Eight global cases as a mechanism spectrum: Station F (flagship by private capital) / Kendall (inclusionary zoning) / one-north (tiered rents) / King's Cross (rail heritage grows an AI cluster) / Pangyo (land for commitments) / Hetao (data-flow pilot) / Toronto Quayside (trust-collapse caution) / Kashiwa-no-ha (tri-party governance) — each annotated with transferable mechanisms and non-transferable premises

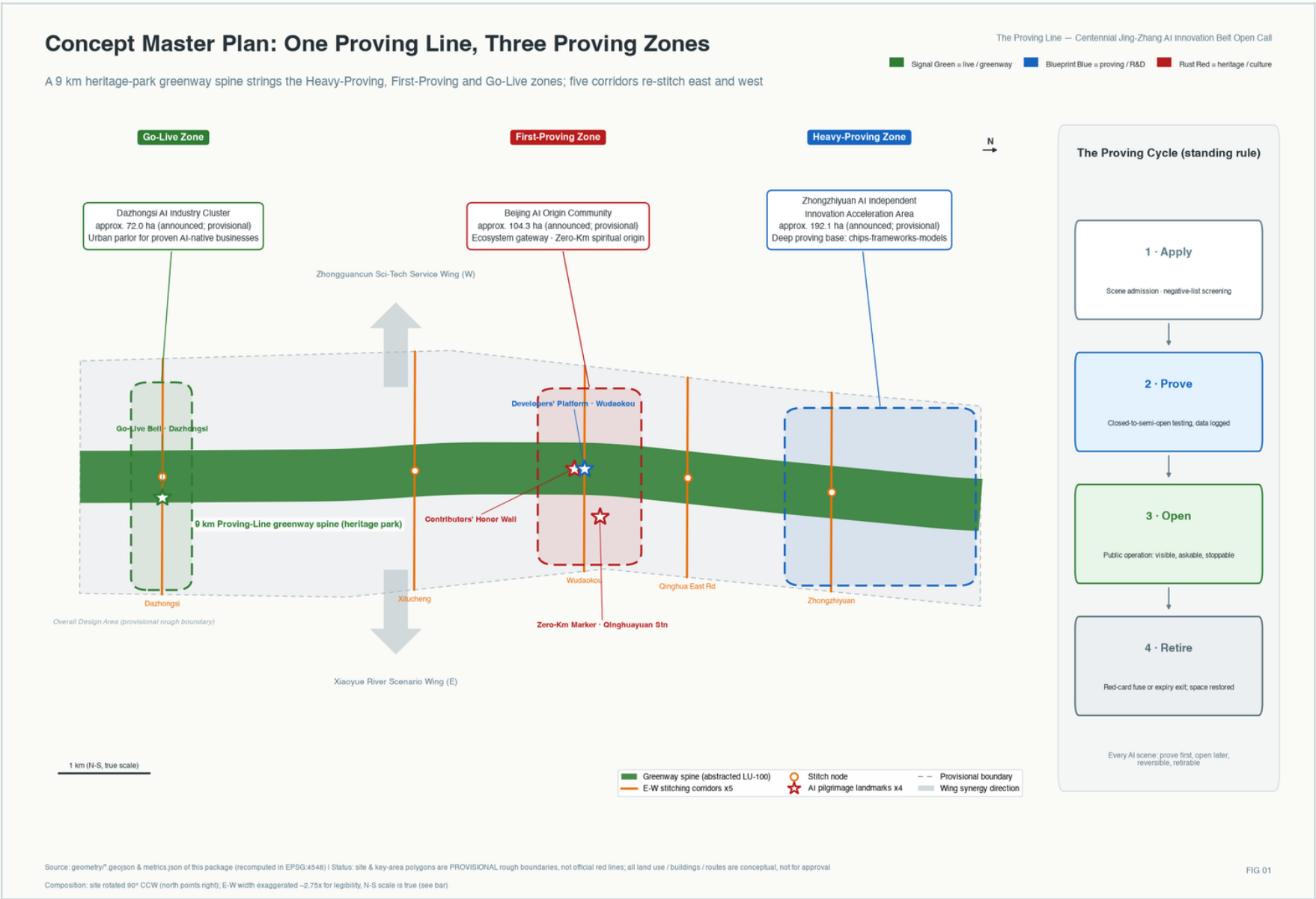
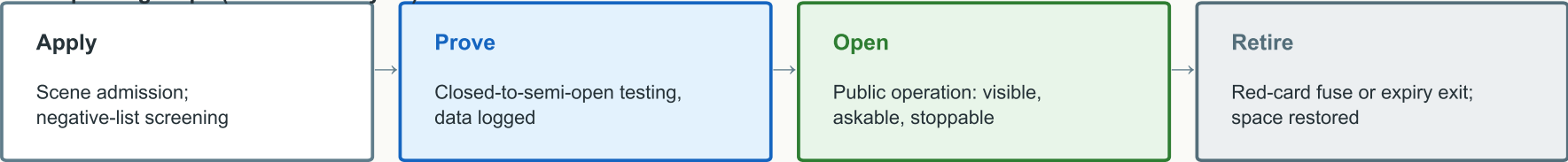


Fig.: one line, three proving districts (from assets/figures/)

Four proving steps (scenario lifecycle)



Land-Use Layout: One Spine, Two Bands

- The central greenway runs the spine; the west band carries research and education spillover; the east band grades through commercial, residential and reserved land
- Codes follow the MNR classification guide; 21 parcels partition the provisional boundary exactly — no gaps, no overlaps (recomputable from geometry and metrics)
- The south strengthens business and culture; the north reserves flexibility — reserved land (16) is the evolutionary testbed

Urban Renewal Framework (conceptual)

- Principle: activate existing stock first; demolition and new build are exceptions
- Three objects: retail-stock renewal (Dazhongsi); campus-edge opening along the five corridors; operations renewal along the park (construction to scenario operations)
- Supply: study a Kendall-style inclusionary innovation-space tool (to be studied)

Retain-Renovate-Demolish (principled classes, conceptual)

- Retain: all heritage assets (incl. Qinghuayuan Station), the completed park, and sound research & housing stock
- Renovate: retail stock (led by Dazhongsi), corridor-edge campus buildings, aged office carriers
- Demolish: only unsafe or gravely inefficient stock confirmed by professional assessment — no target named; parcel-level conclusions belong to statutory procedure

The Four Required Renewal Deliverables, Mapped

Renewal key-area distribution map = the three phasing layers; renewal structure map = one spine, two bands, five stitches; land-use layout map = the 21-parcel land-use layer; renewal project list = the 13 conceptual projects (P11 and geometry/phasing.geojson).

Land-Use Legend (MNR codes)

1401 Park Green Space	0805 Sports
1402 Protective Green Belt	0806 Health Care
0802 Research (R&D)	0701 Urban Residential
0803 Cultural Facilities	05 Commercial & Business
0804 Education	16 Reserved White Land

Formula-Only Discipline

Official controls (FAR, height, coverage, green ratio, setbacks) are unpublished: floor_area_ratio stays unknown in metrics.json with the reason logged; the text gives only formulas and the triggers for assignment. No number is invented — a compliance outcome, not an omission. Total floor area = retained + renewed + new (new build bound by future controls); before official data, any total would be unverifiable pseudo-precision. Fifteen indicative footprints (58245.632 m2 combined) express layout intent only.

Character keynote (conceptual): a modest background city with a legible proving foreground — low-contrast street walls, tricolor semantics at greenway nodes; heights and massing must not overpower the heritage park nor block protected sightlines, with values set by professionals once official controls arrive.

Rail & Station-City Integration

- Line 13 split advancing; Line 18 (Malianwa-Tiantongyuan East) in trial operation since 2025-12-27; Changping extension through the Xueyuan Rd corridor since Dec 2024
- Reinforced connectors at Dazhongsi, Zhichunlu and Wudaokou plus TSI studies (no rail alignment or bridge/tunnel conclusions)

Slow Mobility: 1 Spine + 2 Sides + 5 Cross

- Atop the completed three-path-one-green base: the greenway spine, two cycle side lines and five stitching corridors — 32397.411 m in total (recomputed from layers)
- Corridors first fix crossings over rail heritage and campus walls; alignments conceptual, sections pending professional design
- Parking: transit-first, with proving scenarios self-provisioned (in principle)

Municipal & New Infrastructure

- Distributed edge-compute nodes and charging along the line for near-side proving loads
- Study district micro-grid and compute-load co-dispatch after Kashiwa-no-ha (cited data background only); no utility relocation or capacity conclusions
- Public services for 450,000 residents are the public-interest base: health kiosks, youth third places, plazas doubling as community grounds; capacity needs pending verification

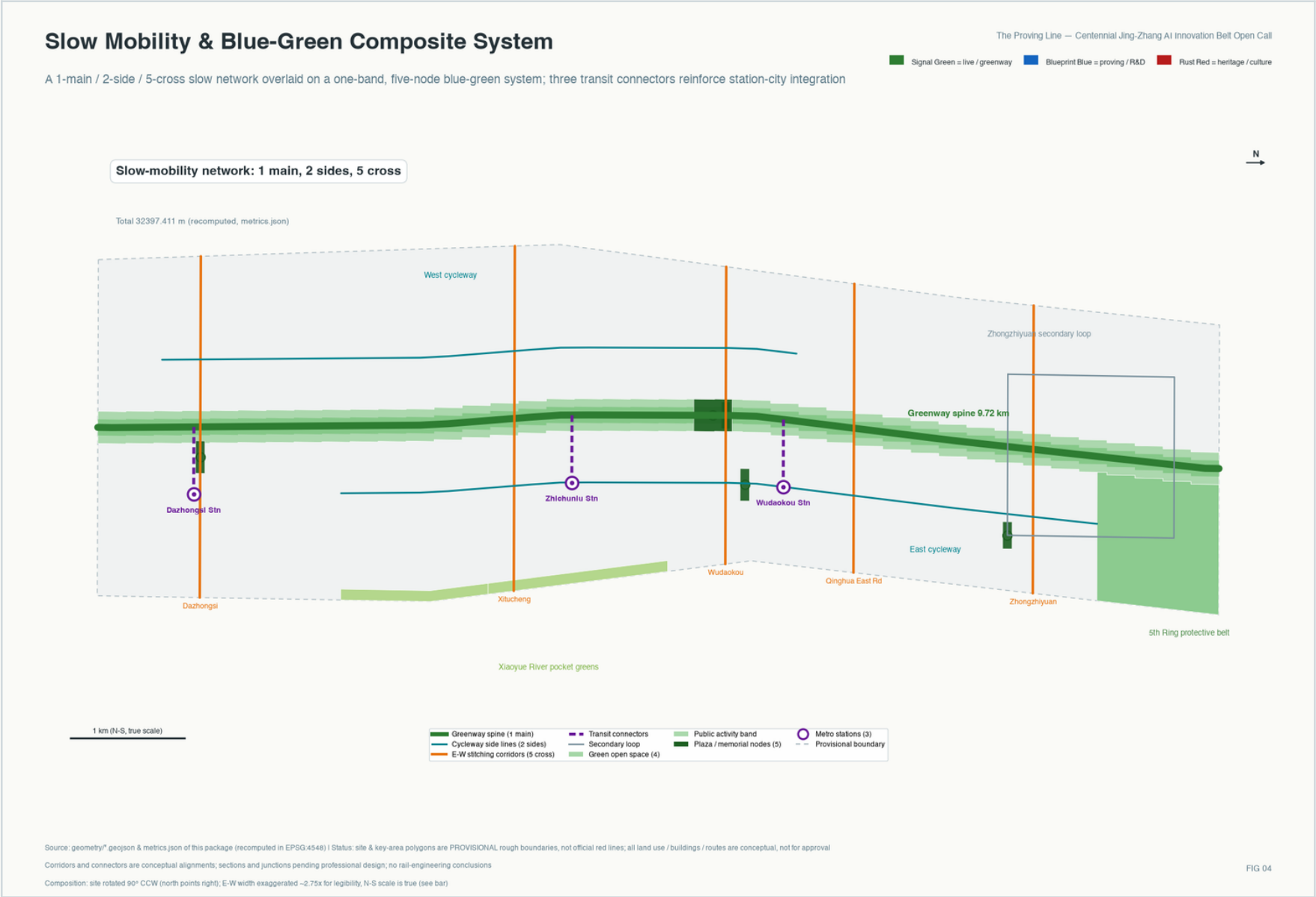


Fig.: slow-mobility network & public activity band (from assets/figures/)

Blue-Green Network & Public Space

- One spine, one belt, many points: the greenway spine meets the 5th-Ring protective belt (N) with two pocket belts along the Xiaoyue River (E); the river itself is locked as-is — no water-side engineering
- Green & open space 1778616.443 m2, green ratio 0.155844 (proposed); public space 628073.744 m2, ratio 0.055032 (union basis, proposed)
- One public activity band + five nodes: station forecourt, Wudaokou plaza, Go-Live Plaza, Launch Plaza, honor-wall band

Four AI Pilgrimage Landmarks (conceptual; rights & deepening required)

- **Zero-Km Marker**
Qinghuayuan Station forecourt: century-old acceptance origin overlaps the AI go-live origin
- **Contributors' Honor Wall**
Engraves proven schemes and their human & agent contributors, linked to this open call's memorial system
- **Go-Live Bell**
Dazhongsi Go-Live Plaza: each proven scenario rings the bell, echoing Juesheng Temple's bell culture
- **Developers' Platform**
Wudaokou: youth gathering place shaped by platform & herringbone canopy imagery

Ten Standardized Components

- Proving milestone
- Scenario status post
- Rail-spike bench
- Station-sign wayfinding
- Reserved-frame device
- Signal mast
- Privacy notice
- Rail planter
- Check-mark strip
- Mobile pod

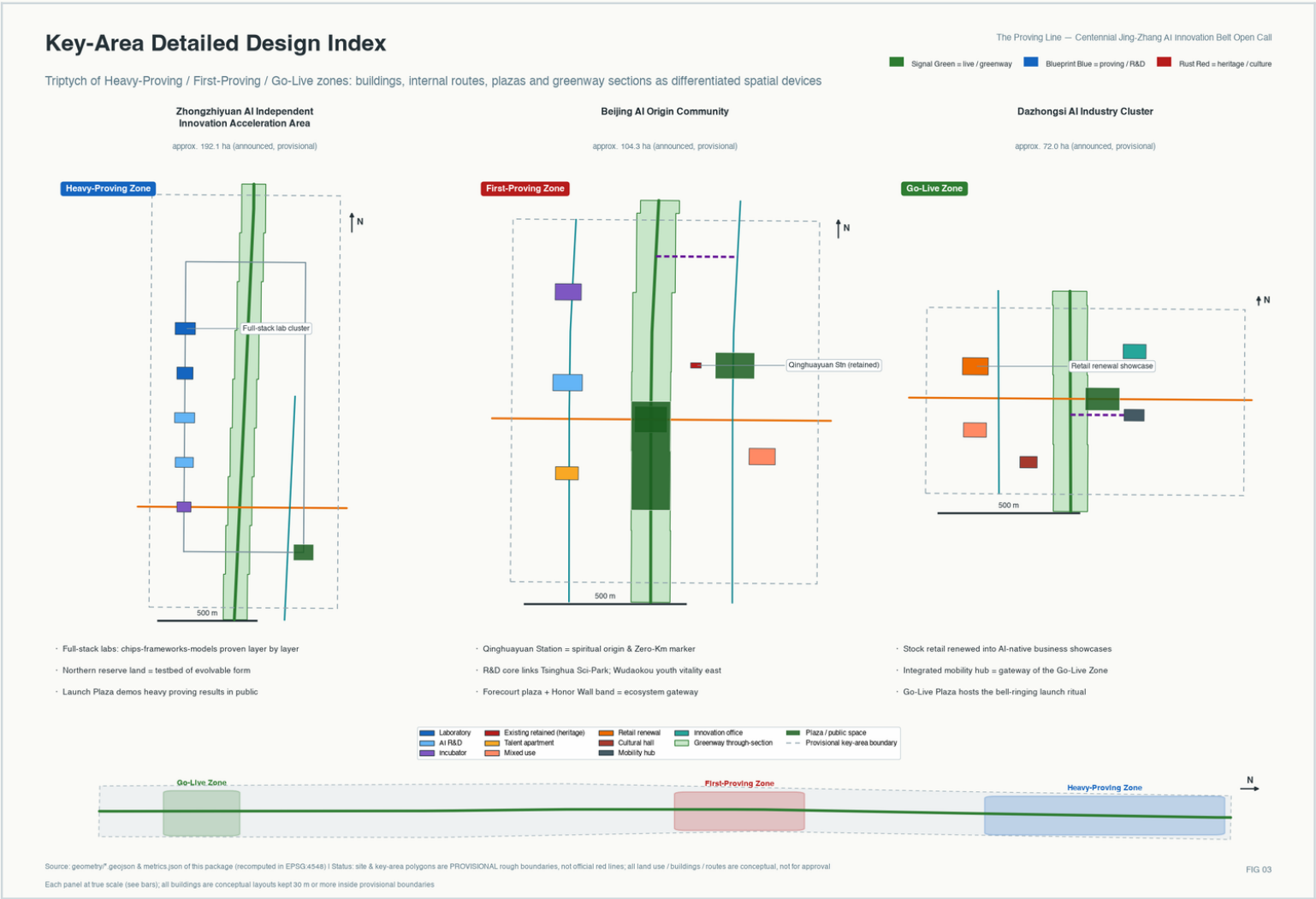
Character & Narrative

- Keynote: a modest background city, a legible proving foreground — low-contrast street walls (grey brick, warm white), tricolor semantic nodes; heights must not overpower the park nor block heritage sightlines; values await official controls
- Three cultural strata: centennial Jing-Zhang (the herringbone alignment lies in Changping-Yanqing as national heritage — honored by narrative, not occupation) + Zhongguancun innovation + the new AI culture of proof before opening
- Three honor tiers: wall (permanent) / milestone columns (annual) / digital twin (real-time); mileposts carry dual 1909 and 2026 scales
- International narrative: a century ago Chinese engineers earned respect through acceptance here; today global AI earns the city's trust through proof

Each key area forms a complete sub-scheme — positioning, spatial structure, building renewal, mobility, public space, AI scenarios and implementation risks — built entirely on provisional rough boundaries as directional detailed design (conceptual). Together c.368.4 ha at implementation-plan design depth.

Role colors: Blueprint Blue = Heavy-Proving (in proof), Rust Red = First-Proving (historic origin), Signal Green = Go-Live (live) — consistent with the tricolor semantics.

Fig.: key-area detailed design index (from assets/figures/)



Heavy-Proving Zone

Zhongzhiyuan AI Independent Innovation Acceleration Area · c.192.1 ha

- Positioning: deep proving base for the full-stack AI system (chips-frameworks-models-agents heavy testing), cultivated toward a national-level AI cluster (conceptual)
- Structure: full-stack lab cluster (S) / shared sports & living (M) / long-term reserve + 5th-ring protective belt (N)
- Reserve = evolutionary-form testbed: only proven functions solidify
- Mobility: secondary loop for freight & commuting; corridor to Qinghe
- Public space: Launch Plaza demos heavy proving results
- Risk: northern baseline missing; recalibrate after official boundary

First-Proving Zone

Beijing AI Origin Community · c.104.3 ha

- Positioning: the ecosystem gateway for global AI talent & projects (300+ AI firms, over 70% share, in the Dongsheng core space)
- Structure: R&D core (W) toward TusPark / Wudaokou youth retail (E)
- Spiritual origin: Qinghuayuan Station (built 1910, heritage, retained)
- Renewal: incubators, talent housing, mixed-use via retrofit + limited new build (conceptual)
- Public space: forecourt plaza (Zero-Km Marker) + honor-wall band
- Risk: complex tenure; renewal sequence verified building by building

Go-Live Zone

Dazhongsi AI Industry Cluster · c.72.0 ha

- Positioning: urban parlor where proven AI-native businesses (consumption and business alike) operate at scale for the public
- Structure: AI-native consumption (W) / industry carrier renewal (E)
- Renewal: closed-for-renovation retail (since 2025) becomes the AI-native demonstration carrier
- Mobility: integrated station hub (conceptual; no engineering claim)
- Public space: Go-Live Plaza — bell-ringing narrative with Juesheng Temple
- Risk: heritage-adjacent character and renewal pacing need professional coordination

Thirteen scenario cards (four industrial testing & validation cards, starred): each states spatial seat, served personas, privacy bounds, human review, operator type, cost structure (composition only), red-card pause condition and post-exit disposition — answerable to 'who is accountable, how does it stop, what happens to the space'. All conceptual.

#	Scenario	Seat	Operator type	Red-card condition
1	* Low-speed robot delivery segment	South service lane + Dazhongsi corridor	Licensed logistics + sub-district oversight	Any collision, or 3 days of complaints over threshold -> pause
2	* Autonomous low-speed shuttle loop	Zhongzhiyuan secondary loop	District platform entity	Beyond safety-operator takeover -> stop
3	* AI health-service navigation kiosk	Xueyuan Rd health-service area	Community health institution	Any clinical misdirection -> offline & fix
4	* Urban-agent adversarial test field	Zhongzhiyuan full-stack lab zone	Professional evaluation body	Real personal data on site -> end batch
5	AI cultural guide	Qinghuayuan Station + full line	Park operator + heritage bodies	Verified factual error -> content offline
6	AI-assisted walkability assessment	Five corridors + Xiaoyue riverside	Transport institute + sub-district	Re-identifiable trajectories -> rectify
7	Enterprise-service copilot	First-Proving R&D core	Park service platform	Reply taken as admin promise -> rectify
8	Public-safety operations review	Temporary command during large events	Local administrative authority	Any automated disposition -> deactivate
9	Developers' Platform (3rd place)	Wudaokou Proving Plaza	Social operator	Encroach public hours -> rectify
10	AI-native retail demonstration	Dazhongsi renewed retail carrier	Commercial operator	Forced biometrics / no cash -> rectify
11	Honor wall & contributor memorial	Honor-wall memorial band	Belt operating agency	Unauthorized attribution -> remove
12	Digital-twin public display	Three proving-district plazas	Belt agency + district platform co.	Data stale > 2 weeks -> offline
13	AI expo & proving-showcase hall (AI-native business)	Dazhongsi east renewal carrier	Commercial operator + belt agency (admin-committee liaison)	False publicity or unauthorized data display -> rectify

Industrial testing: cards 1-4

Closed to semi-open, seated in the Heavy-Proving zone and service lanes

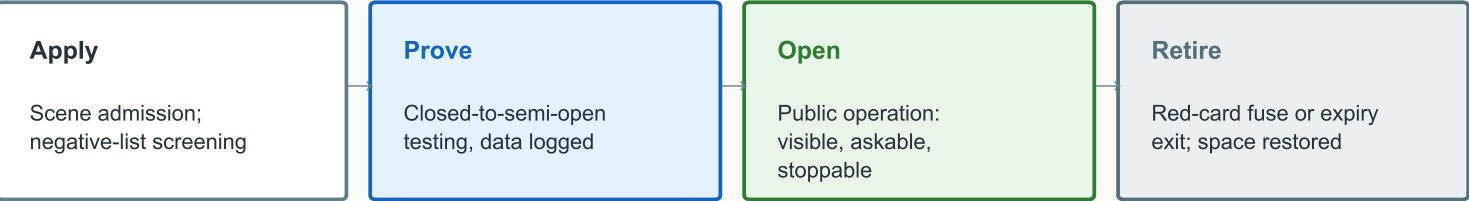
Public-service augmentation: cards 5-7

Along the line, the corridors and the east wing

Public space & operations: cards 8-13

Anchored to the plazas, honor band and Go-Live carriers

The Four-Step Proving Rule (standing)



Uniform privacy floor: anonymous by default, minimal collection, public-sector data governance, accountable human review — the one-vote veto distilled from Toronto Quayside's failure

Annual Programme

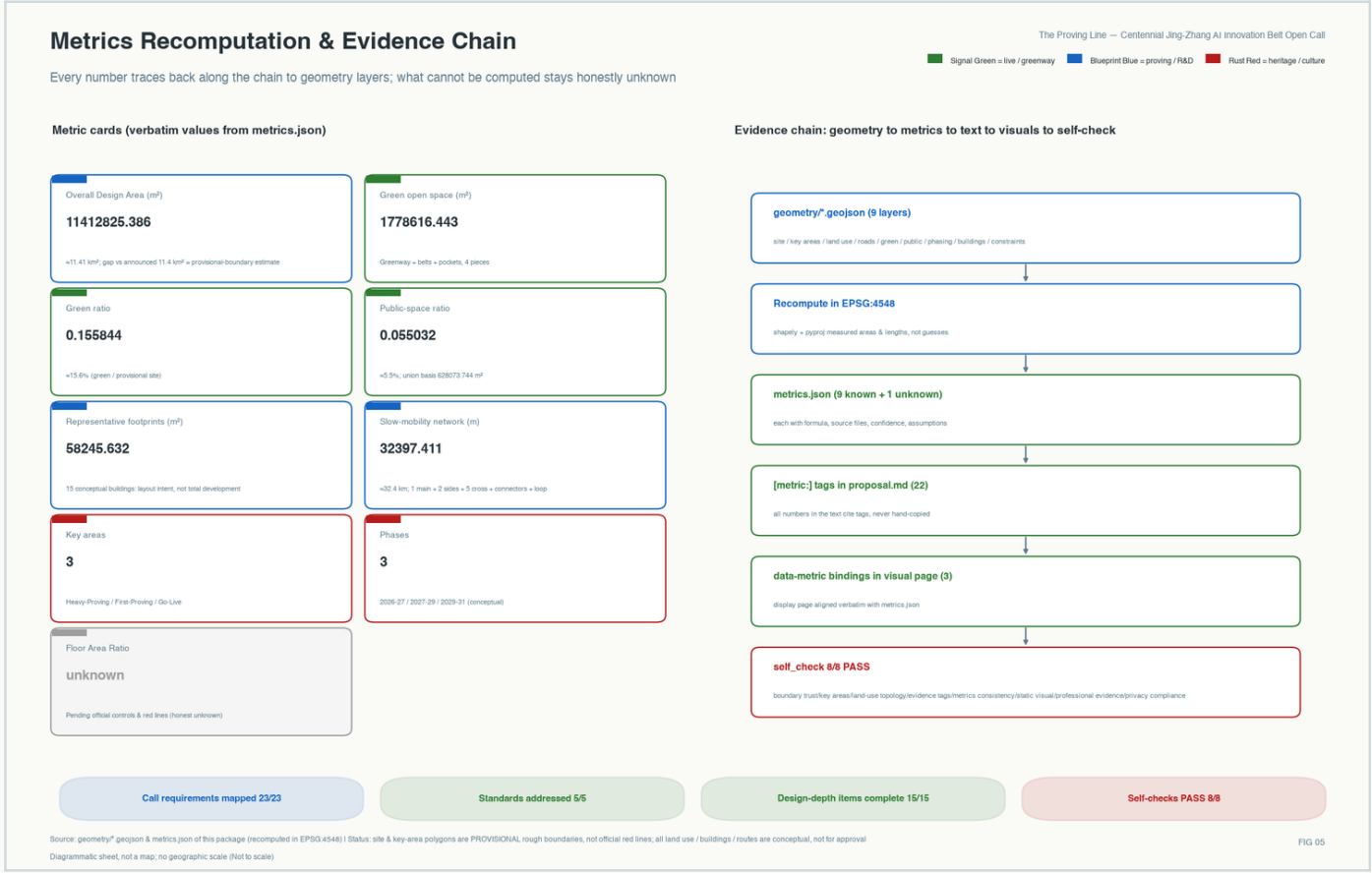
- Spring · Line-Entry Day: proving applications open
- Summer · Go-Live Festival: bell ceremony & public open week at Dazhongsi
- Autumn · Global Agent Urban-Design Open: annualized continuation of this call (conceptual)
- Winter · Proving Review: public replay of the year's operating data

Five personas: AI developers & startups / faculty & students / residents along the line (c.450,000) / enterprises & investors / visitors & international peers — each mapped into the scenario cards

Structural Metrics (EPSG:4548, verbatim from metrics.json)

Metric	Value	Unit	Note
Overall Design Area (derived)	11412825.386	m²	announced 11.4 km² prevails
Green & open space area	1778616.443	m²	proposed value
Green ratio (proposed)	0.155844	-	derived from provisional boundary
Public space area (union)	628073.744	m²	proposed value
Public-space ratio (proposed)	0.055032	-	derived from provisional boundary
Slow-mobility network length	32397.411	m	1 spine + 2 sides + 5 cross
Footprints of 15 indicative buildings	58245.632	m²	layout intent only
FAR: unknown (no official controls)	—	-	formula-only discipline
Key detailed-design areas	3	no.	
Conceptual phases	3	no.	

Area discipline: formal areas follow the announcement (43.6 km2 / 11.4 km2 / 368.4 ha and the three key areas); self-computed areas serve internal consistency only, flagged as derived. For announcement-named indices (AI innovation index, talent density, output scale, intensity), the formula-only rule applies: dimensions and formulas are given, values await official statistics and controls.



Eight-Dimension Risk Matrix (of 5; see risk.json)

Dimension	Score	Mitigation
Data privacy	4	Anonymous by default, minimal collection, public-sector governance; Quayside lesson = first commandment
Implementation complexity	4	Phased delivery, operations before construction, 'three-plus-one' platform consultation
Policy uncertainty	4	Formula-only discipline; every conclusion flagged pending official data
Public acceptance	3	Red-card fuse and public veto up front, humanly received by sub-district and community councils; stale twin goes offline
Spatial disputes	3	No demolition targets named; conservative near heritage; routes TBD
Technology maturity	3	Proving follows industry test standards with third-party human-signed reports (card 4), never this scheme's own flow alone
Operating cost	3	Cost structure disclosed per card; retirement prevents zombie assets
Equity & inclusion	2	Non-AI alternatives, accessibility and cash/human channels mandatory

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Compliance Boundary Restated

Everything here is an open co-creation concept — it does not replace statutory planning nor constitute an approved government conclusion; statements touching regulatory adjustment, FAR, heights, retain-renovate-demolish, road alignments, utilities, investment or sequencing are research suggestions pending official confirmation; all events, policies and operating arrangements are envisioned mechanisms binding no party.

Nine Pending-Data Classes (see assumptions.json)

- Official precise red line (ODA)
- Official polygons of the three key areas
- Official regulatory controls (FAR / height / coverage / setbacks)
- Road red lines, sections, junctions
- Parcel tenure and ownership baseline
- Building stock baseline
- Heritage protection & control-zone GIS lines
- Municipal utility baseline & capacities
- Public-service capacity data

Filling any item triggers recomputation of the affected chapters and layers — the gap register is not a disclaimer but the task list for the next iteration.

Public Red-Card Procedure (conceptual)

- Complaints received via the digital twin and offline points
- Joint petitions or complaint volume at the pre-published threshold trigger human review
- Review conclusions published within 7 working days — the procedure itself is proved first in the First-Proving zone; the 'three-plus-one' governance adds a rotating community seat for the 70 communities